Curriculum Vitae – Anders Stevnhoved Olsen

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Current appointments

2025 –	Postdoc, Neurobiology research unit, Rigshospitalet, Copenhagen, DK. Mentor: Prof. Gitte M Knudsen
2026	Postdoc (3-month secondment), Neuro-ICU, Rigshospitalet, Copenhagen, DK. Mentor: Prof. Daniel Kondziella

Previous appointments (research related)

2024	Research assistant (3-month PhD secondment), DTU Compute, DK. Supervisor: Bjørn S Jensen.
2021	Research assistant, Neurobiology Research Unit, Rigshospitalet, Copenhagen, DK.
2019 – 2021	Pre-graduate research assistant, Neurobiology Research Unit, Rigshospitalet, Copenhagen, DK (part-time).

Education

2021 – 2025	PhD student, Section for Cognitive Systems, DTU Compute, Technical University of Denmark. Project title: *Uncovering Brain Dynamics using Directional Statistics and Functional Neuroimaging Data.* Supervisors: Prof. Morten Mørup, prof. Kristoffer H Madsen, assoc. prof. Patrick M Fisher.
2023	Visiting PhD student (4-month secondment), Ecole Polytechnique Fédérale de Lausanne, Geneva, CH.
2019	Exchange student (MSc) at Ecole Polytechnique Fédérale de Lausanne, Lausanne, CH
2015 – 2020	BSc & MSc in Biomedical Engineering, Technical University of Denmark & University of Copenhagen

Scientific focus area

My research centres on developing mathematically sound and computationally useful tools for functional neuroimaging analysis. My work lies at the intersection of unsupervised machine learning, directional statistics, signal processing, and neuroscience, with applications across fMRI, EEG/MEG, PET, and SPECT including dynamic brain connectivity, consciousness and altered brain states, brain structure–function relationships, and glymphatic physiology. I have first-authored work in PNAS, Nature Communications, NeuroImage, and Frontiers in Neuroscience, alongside co-author publications in leading neuroscience journals and several contributions to IEEE conference proceedings.

Selected research projects

⊛ **Directional statistics and brain dynamics**: The paper *Uncovering dynamic human brain phase coherence networks* is in press in PNAS [1], and I am currently working on an IEEE overview of directional statistics in signal processing with Profs. Kanti Mardia, Dimitri van de Ville, and Morten Mørup.
⊛ **Multimodal EEG/MEG modeling**: My works in Frontiers in Neuroscience and EUSIPCO developed unsupervised approaches for identifying shared brain dynamics while accounting for variation across subjects and imaging modalities.
⊛ **Psychedelics and consciousness**: A Nature Communications [2] and NeuroImage paper [12] investigate how psilocybin alters dynamic connectivity and brain entropy, and a preprint investigates frequency-specific fMRI effects of psilocybin [3].
⊛ **Disorders of consciousness**: In an ongoing project, I investigate the effects of psilocybin on the comatose patients in the neuro-ICU measured using EEG with prof. Daniel Kondziella. A student of mine is investigating the dynamics of brain connectivity in fMRI data of anesthetized healthy subjects.
⊛ **Glymphatic physiology**: A paper in PLOS Biology [6] examines how sleep, cerebral vasomotion, and respiration- and cardiac-driven brain pulsations may contribute to cerebrospinal-fluid movement and brain waste clearance. Ongoing work include SPECT in porcine models [5] and human normal pressure hydrocephalus / idiopathic intracranial hypertension.
⊛ **Structure-function**: Two projects with prof Dimitri van de Ville and Harry H. Behjat examine how cortical geometry and structural connectivity (don't) constrain functional brain maps [7,8].

Supervision, teaching, funding, and peer reviews

⊛Supervised 4 MSc theses, 5 BSc theses (6 separate students), 10 semester project groups (~30 separate students). ⊛TA'ed introductory data science methods (3 semesters) and introductory machine learning (3 semesters). ⊛Recipient of the highly competitive 3-year DTU Compute PhD scholarship. ⊛Peer-reviewed for NeuroImage, Network Neuroscience, Frontiers in Psychiatry, Communications Biology x4, Nature Mental Health, Annals of NY academy of science.

Conference attendance ([1]poster, [2]oral presentation)

OHBM 2021[1] (online), 2022[1] (Glasgow), 2024[1] (Seoul), 2026[1,2] (Bordeaux), ICPR 2022 (Haarlem), DDS 2022[1] (Billund), ICASSP 2023[1] (Rhodes), 2026[1] (Barcelona), Resting state 2023[1] (Dallas), EUSIPCO 2024[1] (Lyon), ISBI 2025[1,2] (Houston).

Volunteering

2018	Medical technician and hygiene educator in Nepal, Engineering World Health
2017 – 2018	Fundraising coordinator, Engineering World Health Denmark

Previous appointments (not research related, all part-time during BSc/MSC studies)

2017 – 2019	Data analysis assistant, CIMT, Capital region of Denmark
2017 – 2018	Instructor in the course Cell- and Tissue Biology, Faculty of Health at University of Copenhagen
2017	Student assistant, Center for Hyperpolarization in Magnetic Resonance, Technical University of Denmark
2017	IT specialist, Diagnostic center, Rigshospitalet

List of publications (†first author). H-index: 6. Abstracts not included.

1. **AS Olsen**†, A Brammer, PM Fisher, M Mørup (2026): *Uncovering dynamic human brain phase coherence networks* (PNAS, in press).
2. DE McCulloch†, **AS Olsen**†, B Ozenne, DS Stenbæk, S Armand, MK Madsen, GM Knudsen, PM Fisher (2026): *Multi-metric evaluations of acute psychedelic effects on fMRI brain entropy*. (Nature Communications).
3. **AS Olsen**†, K Larsen, DE McCulloch, M Ganz, MK Madsen, B Ozenne, GM Knudsen, N ur Rehman, PM Fisher (2026): *Psilocybin acutely reduces low-frequency BOLD power and frequency-specific connectivity* (bioRxiv, in review)
4. K Larsen†, DE McCulloch, **AS Olsen**, F Müller, ME Liechti, S Borgwadt, F Holze, L Ley, P Vizeli, A Klaiber, AM Becker, B Ozenne, M Avram, PM Fisher (2026): *Psychedelics distinctly alter brain entropy and complexity compared to psychostimulants* (medRxiv, in review).
5. **AS Olsen**†, ML Navarro, C Svarer, JL Hinrich, M Mørup, GM Knudsen (2026): *Shift- and stretch-invariant non-negative matrix factorization with an application to brain tissue delineation in emission tomography data* (ICASSP).
6. SMU Larsen†, SC Holst†, **AS Olsen**, B Ozenne, DB Zilstorff, K Brendstrup-Brix, P Weikop, S Pleinert, V Kiviniemi, PJ Jennum, M Nedergaard, GM Knudsen (2025): *Sleep deprivation promotes cerebral vasomotion while respiration- and cardiac-driven brain pulsations escalate with sleep intensity*. PLoS Biology.
7. A Milloz†, J Vogel, **AS Olsen**, JC Pang, O Strandberg, TE Anijärv, S Palmqvist, N Sportono, R Ossenkoppele, D Van De Ville, O Hansson, H Behjat (2025): *Multiscale Quantification of hemispheric asymmetry in Cortical Maps using Geometric Eigenmodes* (ISBI).
8. **AS Olsen**†, S Mansour L, JC Pang, A Zalesky, D Van De Ville, H Behjat (2025): *On reconstruction of cortical functional maps using subject-specific geometric and connectome eigenmodes* (ISBI).
9. **AS Olsen**†, Nielsen JD, Mørup M (2024): *Coupled generator decomposition for fusion of electro- and magneto-encephalography data*. EUSIPCO.
10. NB Iskov†, **AS Olsen**, KH Madsen, M Mørup (2024): *Discovering Prominent Differences in Structural and Functional Connectomes Using a Multinomial Stochastic Block Model.* Network Neuroscience.
11. **AS Olsen**†, E Ortvald, KH Madsen, MN Schmidt, M Mørup (2023): *Angular central Gaussian and Watson mixture models for assessing dynamic functional brain connectivity during a motor task*. ICASSP.
12. **AS Olsen**†, A Lykkebo-Valløe, B Ozenne, MK Madsen, DS Stenbæk, S Armand, M Mørup, GM Knudsen, PM Fisher (2022): *Psilocybin modulation of dynamic functional connectivity is associated with plasma psilocin and subjective effects*. Neuroimage.
13. **AS Olsen**†, RMT Høegh†, JL Hinrich, KH Madsen, M Mørup (2022): *Combining electro-and magnetoencephalography data using directional archetypal analysis*. Frontiers in Neuroscience.
14. KHR Jensen†, DE McCulloch†, **AS Olsen**†, SEP Bruzzone, SV Larsen, PM Fisher, VG Frøkjær (2022): *Effects of an Oral Contraceptive on Dynamic Brain States and Network Modularity in a Serial Single-Subject Study*. Frontiers in Neuroscience.